

TROITSKO-PECHORSK, Russian Federation

WMO: 237110

Lat: 62.7324N

Lon: 56.1930E

Elev: 446

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.3	(c) -28.5	(d) -39.8	(e) 0.6	(f) -34.3	(g) -34.3	(h) 0.8	(i) -28.5	(j) 14.5	(k) 15.5	(l) 12.7	(m) 14.5	(n) 1.5	(o) 180	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.8	(c) 83.5	(d) 67.0	(e) 79.3	(f) 64.8	(g) 75.2	(h) 62.4	(i) 68.7	(j) 79.6	(k) 66.6	(l) 76.4	(m) 64.4	(n) 72.4	(o) 4.9	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.3	73.4	62.8	86.9	71.2	60.8	80.9	69.5	33.1	80.3	31.4	77.0	29.7	72.5	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.2	11.7	10.1	-37.7	88.1	7.3	3.1	-42.9	90.3	-47.2	92.1	-51.2	93.8	-56.5	96.0		
			-36.9	71.6	7.8	2.2	-42.5	73.1	-47.1	74.4	-51.5	75.7	-57.1	77.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	1.6	4.5	19.3	31.8	43.6	55.1	61.5	54.8	45.0	32.9	17.1	7.6		
	DBStd	23.29	16.13	15.52	10.86	9.90	9.54	9.28	8.59	7.62	7.11	8.29	13.96	17.50		
	HDD50	7628	1499	1273	951	550	248	50	11	34	181	530	987	1314		
	HDD65	12364	1964	1693	1416	997	664	319	172	329	600	994	1437	1779		
	CDD50	843	0	0	0	3	51	204	368	184	32	1	0	0		
	CDD65	103	0	0	0	0	2	23	64	14	0	0	0	0		
	CDH74	1086	0	0	0	0	44	263	650	128	1	0	0	0		
	CDH80	282	0	0	0	0	7	64	178	33	0	0	0	0		
(14) Wind	WSAvg	5.1	4.7	4.9	5.7	5.8	5.7	5.3	4.7	4.7	5.0	5.4	5.0	4.9		
(15) Precipitation	PrecAvg	24.9	1.7	1.2	1.1	1.5	2.1	2.4	2.8	3.1	2.4	2.6	2.2	1.9		
	PrecMax	32.1	2.6	2.4	2.9	4.3	5.7	6.5	5.7	5.3	4.3	4.8	3.5	3.5		
	PrecMin	20.4	0.7	0.1	0.3	0.6	0.4	0.6	0.5	0.7	0.7	0.6	0.6	0.6		
	PrecStd	2.8	0.4	0.5	0.6	0.7	1.0	1.2	1.2	1.2	0.9	1.0	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	34.4	44.9	64.9	80.5	86.2	88.5	84.7	68.7	55.9	42.9	36.1		
		MCWB	31.7	32.6	36.0	50.9	60.3	66.5	68.8	68.1	58.7	50.4	42.1	35.5		
	2%	DB	29.5	31.7	39.4	55.0	73.1	80.8	84.9	76.9	63.0	51.3	37.6	32.9		
		MCWB	28.5	30.4	33.2	44.6	56.1	64.6	67.8	64.4	54.7	48.4	36.5	32.1		
	5%	DB	26.5	29.1	36.1	50.0	66.7	76.1	81.3	72.0	59.3	47.4	34.6	31.0		
		MCWB	25.6	27.9	32.2	40.8	53.0	62.2	66.4	61.2	53.4	44.8	33.8	30.3		
	10%	DB	22.6	25.8	33.5	45.6	61.3	71.2	77.3	67.7	55.8	44.3	32.8	28.3		
		MCWB	21.7	24.6	30.8	38.4	50.2	59.3	64.5	59.7	51.4	42.4	32.1	27.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	33.1	37.5	51.3	61.0	69.8	71.9	69.2	60.1	52.0	42.1	35.0		
		MCDB	32.1	33.8	42.7	63.0	75.6	80.6	84.1	81.6	67.0	54.5	42.8	35.5		
	2%	WB	28.8	30.7	34.6	45.1	57.9	66.6	69.7	66.4	56.3	48.1	36.5	32.3		
		MCDB	29.6	31.7	37.7	53.7	70.6	77.7	80.6	75.0	60.7	50.6	37.3	32.7		
	5%	WB	25.7	27.9	32.9	41.9	54.5	63.8	67.6	63.2	54.1	45.1	34.0	30.3		
		MCDB	26.6	29.0	35.5	47.9	64.5	73.3	77.9	68.9	58.3	47.3	34.4	31.0		
	10%	WB	21.8	24.8	31.0	39.3	51.3	61.0	65.9	60.6	51.8	42.4	32.3	27.5		
		MCDB	22.6	25.9	33.6	44.5	60.1	69.5	75.2	66.1	55.4	44.1	32.8	28.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	13.7	16.3	16.4	18.0	18.1	18.8	15.5	12.3	8.0	9.1	10.8		
		MCDBR	12.8	9.1	15.1	22.9	28.7	24.9	24.9	23.5	18.1	12.1	6.9	7.4		
	5% WB	MCWBR	12.5	8.6	11.2	14.6	15.4	11.7	10.5	11.6	10.7	9.1	7.0	7.0		
		MCWBR	12.5	8.7	12.3	20.2	25.5	23.1	22.3	20.0	15.7	11.5	7.0	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.294	0.331	0.358	0.360	0.366	0.403	0.381	0.340	0.312	0.249	0.234		
		taud	1.938	2.025	2.027	2.084	2.374	2.439	2.337	2.375	2.458	2.318	2.013	1.911		
	Ebn at Noon		114	195	236	258	272	272	256	252	242	197	113	56		
		Edh at Noon	12	23	35	40	33	32	35	30	23	18	10	6		
(44) All-Sky Solar Radiation	RadAvg		74	305	738	1253	1576	1621	1645	1110	623	233	78	33		
	RadStd		7	44	83	128	129	118	154	139	83	30	17	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+1.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page